Merged Components List

October 20, 2026 · Ziad Mostafa

One list for the hospital delivery robot, in the original components-list format, with the power subsystem updated to the 24V scooter-class motor design.

Components

No.	Category	Component	Name / Model to Search	Price (EGP)
1	High-Level Computer	Raspberry Pi 4 Model B 8GB	Raspberry Pi 4 Model B 8GB	9,500
2	Storage	microSD card	SanDisk Extreme microSDXC 128GB	3,200
3	STM32 Development Board	STM32 board	STM32 NUCLEO-F401RE	2,100
4	LiDAR Sensor	2D LiDAR	Slamtec RPLIDAR A1M8	8,500
5	Camera	Pi camera	Raspberry Pi Camera Module 3	4,500
6	Distance Sensor	Time-of-flight sensor	VL53L1X Time-of-Flight Sensor	330
7	IMU Sensor	6-axis IMU	MPU-6050 6-Axis IMU	175
8	Drive Motors and Wheels	2x scooter-class BLDC hub motor, 24V 250W (350W peak), built-in Hall sensors and wheel	24V 250W BLDC Hub Motor, 8-10 inch	1,200–1,800 each
9	Motor Drivers	2x BLDC controller, 24V, 15A or more, Hall input, PWM/DIR/brake	24V 15A BLDC Motor Controller with Hall Input	600–900 each
10	Battery and BMS	LiFePO4 8S pack, 24V 15Ah (about 384 Wh), with BMS (30A continuous / 60A peak)	24V 8S LiFePO4 15Ah Battery Pack with BMS	6,500–9,000
11	Electrical Protection	40A main fuse at battery positive, holder, branch fuses (set)	Automotive Blade Fuse Holder and Fuses, 40A	200–300
12	Voltage Regulation	2x buck converter: 24V to 5V 5A (Pi rail) and 24V to 5V 3A (logic rail)	24V-to-5V 5A DC-DC Buck Converter; 24V-to-5V 3A DC-DC Buck Converter	180–280 and 150–220
13	CAN Transceiver	CAN transceiver	SN65HVD230 CAN Transceiver	850
14	Raspberry Pi CAN Interface	CAN controller	MCP2515 CAN Controller Module	400
15	Emergency Stop and Safety	Mushroom e-stop button, main disconnect switch (40A or more), e-stop relay (40A)	22mm Mushroom Emergency Stop Switch; 40A Automotive Disconnect Switch; DC-Rated Contactor and Interlock Switch	500–800 (set)
16	Robot Chassis	Chassis	4-Wheel Mobile Robot Chassis / Custom Aluminum Chassis	—
17	Delivery Compartment	Lockable box	Lockable Metal Storage Box	—
18	Access Control	RFID reader	RC522 RFID Reader Module	120
19	Fasteners	Screw and nut kit	M3/M4 Stainless Steel Screw and Nut Kit	—
20	Fabrication Materials	Aluminum stock	Aluminum Sheets and 2020 Aluminum Extrusions	—
21	Cooling and Accessories	Fan and heatsink	5V Cooling Fan and Heatsink	—

No.	Category	Component	Name / Model to Search	Price (EGP)
22	Face Display	7-inch display for animated robot face (also serves as the optional touchscreen)	7-Inch HDMI LCD Display for Animated Robot Face	4,000 / 8,000
23	Head Enclosure	Robot head shell	Custom 3D-Printed Robot Head Shell	—
24	Face Camera	USB camera	Wide-Angle USB Camera	1,300
25	Voice Input	Microphone	USB Microphone Module	—
26	Voice Output	Speaker and amplifier	Mini Speaker with Audio Amplifier	—
27	Humanoid Arms	Arms	Custom 3D-Printed Robot Arms	—
28	Torso Structure	Torso	Custom Robot Torso Enclosure	—
29	Neck Structure	Neck bracket	Robot Neck Mounting Bracket	—
30	Head Movement	Pan-tilt servo	Metal Gear Servo Motor with Pan-Tilt Bracket	—
31	Display Mounting	LCD bracket	Custom LCD Mounting Bracket	—
32	Mechanical Assembly	Brackets and hardware	Aluminum Brackets, Hinges, and Mounting Hardware	—
33	Power Distribution	Fused distribution board with star-ground bus	Fused Power Distribution Board	—
34	Charging Dock	Dock charger, 29.2V 5A CC-CV, LiFePO4 profile	29.2V 5A LiFePO4 Charger (8S)	1,200–1,800
35	Power Monitoring	Bus voltage/current sensor, 36V rated	INA226 Voltage Current Sensor Module	150–220
36	Dock Contacts	Spring-loaded pogo pins, Schottky diode (5A or more), 5A fuse, connectors	Pogo Pin Charging Contacts with Schottky Diode and 5A Fuse	350–550 (set)
37	Wiring	Wire 0.35–6 mm^2, terminals, XT60/XT90, heat-shrink	Silicone Wire 6mm 4mm 2.5mm 1mm, XT60 XT90 Connectors	400–650 (set)

Dimensions

Sizes are in cm (L x W x H) and prices in Egyptian pounds (EGP). Rows marked Typical are common sizes for that part type, so confirm against the exact product you buy. Rows marked Proposed are my size estimates for custom or kit parts, to be confirmed once the mechanical design is fixed.

No.	Component	Price (EGP)	Length (cm)	Width (cm)	Height (cm)	Basis
1	Raspberry Pi 4 Model B 8GB	9,500	8.5	5.6	1.7	Retail listing
2	microSD card 128GB	3,200	1.5	1.1	0.1	Standard microSD size
3	STM32 NUCLEO-F401RE	2,100	8.2	7.0	2.0	Store listings (they differ slightly)
4	Slamtec RPLIDAR A1M8	8,500	9.7	7.0	5.5	Manufacturer spec
5	Raspberry Pi Camera Module 3	4,500	2.5	2.4	1.15	Manufacturer spec
6	VL53L1X time-of-flight sensor	330	1.8	1.3	0.3	Typical
7	MPU-6050 IMU	175	2.0	1.6	0.3	Typical
8	24V 250W BLDC hub motor (x2)	1,200–1,800 each	20.0	4.7	20.0	Store listing, 8-inch wheel (diameter 20, width 4.7)
9	BLDC controller 24V 15A (x2)	600–900 each	9.0	5.2	3.0	Store listing, similar 24V 250W controller
10	LiFePO4 8S 24V 15Ah pack with BMS	6,500–9,000	20.0	13.0	12.0	Typical estimate, confirm with seller
11	Fuse holder and fuses (set)	200–300	5.0	2.0	2.0	Typical, inline holder
12a	Buck converter 24V to 5V 5A	180–280	6.0	3.5	1.8	Typical
12b	Buck converter 24V to 5V 3A	150–220	4.3	2.1	1.4	Typical
13	SN65HVD230 CAN transceiver	850	2.9	2.0	0.8	Typical
14	MCP2515 CAN module	400	4.0	2.8	1.3	Typical
15	Emergency stop and safety set	500–800 (set)	4.1	4.1	7.0	Typical, e-stop button only; switch and relay sizes vary by model
16	Robot chassis	—	60.0	45.0	20.0	Proposed size for a 15-25 kg chassis, confirm in mechanical design
17	Lockable delivery box	—	40.0	30.0	30.0	Proposed size, confirm in mechanical design
18	RC522 RFID reader	120	6.0	4.0	0.9	Typical
19	Fasteners kit	—	20.0	12.0	4.0	Typical kit box
20	Aluminum sheets and 2020 extrusions	—	50.0	2.0	2.0	Standard 20x20 mm profile, proposed 50 cm cut length; sheets 60 x 40, 0.3 thick
21	Fan and heatsink	—	4.0	4.0	1.0	Typical
22	7-inch HDMI face display	4,000 / 8,000	16.5	10.0	1.5	Typical
23	Head enclosure	—	25.0	20.0	20.0	Proposed size, confirm in mechanical design
24	Wide-angle USB camera	1,300	3.5	3.5	3.0	Typical
25	USB microphone module	—	6.0	2.0	1.0	Typical

No.	Component	Price (EGP)	Length (cm)	Width (cm)	Height (cm)	Basis
26	Mini speaker with amplifier	—	5.0	5.0	3.0	Typical
27	Humanoid arms	—	35.0	6.0	6.0	Proposed size per arm, confirm in mechanical design
28	Torso enclosure	—	40.0	30.0	45.0	Proposed size, confirm in mechanical design
29	Neck mounting bracket	—	8.0	6.0	10.0	Proposed size, confirm in mechanical design
30	Metal gear servo	—	4.1	2.0	4.3	Typical, servo only; pan-tilt bracket adds size
31	LCD mounting bracket	—	18.0	4.0	12.0	Proposed size, confirm in mechanical design
32	Brackets, hinges, hardware	—	25.0	15.0	6.0	Typical kit box
33	Fused power distribution board	—	10.0	6.0	3.0	Typical
34	Dock charger 29.2V 5A	1,200–1,800	17.0	7.5	4.5	Typical
35	INA226 sensor module	150–220	2.5	2.0	0.4	Typical
36	Dock contacts, diode, fuse (set)	350–550 (set)	8.0	6.0	3.0	Proposed size, confirm in dock design
37	Wiring and terminals (set)	400–650 (set)	15.0	15.0	8.0	Typical packed set

Notes

- **Replaced (12V design):** the 12V geared motors, Cytron MDDS30 driver, 12V LiFePO4 battery and 12V-to-5V buck are replaced by Nos. 8, 9, 10 and 12. The INA219 sensor is replaced by the INA226 (No. 35), since the 8S pack reaches 29.2V.
- **Duplicates merged:** BMS is part of the battery pack (No. 10). The e-stop button is part of the safety set (No. 15). Wheels come with the hub motors (No. 8). The optional touchscreen and the face display were the same 7-inch HDMI screen (No. 22).
- **Added from the power design:** Nos. 34–37 (charger, INA226, dock contacts, wiring).
- **Price basis:** unit price per item; quantities are in the Component column. Ranges are rough local-market figures to confirm with supplier quotes. A dash means no price in either source.
- **Priced subtotal:** the 11 items priced only in the original list total 30,975 EGP, excluding the face display (4,000 / 8,000). The power subsystem totals 13,230–19,200 EGP, so the combined priced total is 44,205–50,175 EGP before the unpriced rows.
- **Check:** hub motors drive two wheels, so the 4-wheel chassis (No. 16) may need passive casters. Hall-only feedback has lower odometry resolution than encoders; add magnetic encoder disks if Nav2 localization suffers.